

1. AI-BASED SMART HOSPITAL EMERGENCY RESPONSE SYSTEM

REAL-TIME CASE STUDY: HOSPITALS AND EMERGENCY DEPARTMENTS.

Problem:

During emergencies, patient data, ambulance availability, doctor availability, ICU beds, and medical devices need to be coordinated instantly.

Novelty:

An AI-driven priority engine continuously analyses patient severity, available resources, and hospital capacity to automatically prioritize emergency cases.

Major Components:

- Patient Management System
- AI Priority Engine
- Emergency Alert Service
- Ambulance Tracking
- Doctor/Staff Management
- Smart ICU Bed Management
- Medical IoT Gateway
- Secure Patient Database
- Notification System

Unique Feature:

If a patient's vital signs suddenly become abnormal, the system can automatically increase emergency priority and notify the appropriate medical team.

2. AI-BASED CYBERCRIME EVIDENCE MANAGEMENT & INVESTIGATION SYSTEM

REAL-TIME CASE STUDY: POLICE/CYBERCRIME INVESTIGATION DEPARTMENTS.

Problem:

Digital evidence such as images, logs, videos, mobile data, and documents can be modified or mishandled during investigations.

Novelty:

A system that combines AI-based evidence classification + blockchain-based integrity verification + forensic chain-of-custody tracking.

Major Components:

- Evidence Collection Module
- AI Evidence Classifier
- Hash Generation Engine
- Blockchain Ledger
- Chain-of-Custody Manager
- Investigator Dashboard
- Evidence Storage
- Audit & Reporting Module

Unique Feature:

Every evidence file receives a cryptographic hash, while every transfer/access event is recorded in an immutable ledger.

Result:

Investigators can verify who accessed the evidence, when it was accessed, and whether the evidence was modified.

3. SMART HOSTEL SECURITY & VISITOR MANAGEMENT SYSTEM

Real-Time Case Study: College hostels.

Problem:

Manual visitor registers cannot effectively verify visitors or track unauthorized access.

Novelty:

A QR + facial verification + anomaly detection system for hostel entry.

Main Classes:

- Student
- Visitor
- Security Guard
- Hostel
- Room
- Visitor Pass
- QR Code

- Face Verification
- Entry Log

Unique Feature:

If a visitor repeatedly attempts to enter using different student details, the system identifies the behaviour as suspicious and generates an alert.

4. AI-BASED PHISHING EMAIL DETECTION SYSTEM

REAL-TIME CASE STUDY: CORPORATE EMAIL SECURITY.

Problem:

Employees receive phishing emails that attempt to steal credentials or sensitive information.

Novelty:

Instead of checking only suspicious keywords, the system analyses sender identity, URL characteristics, language patterns, domain reputation, and email behaviour.

Main Classes:

- Email
- Sender
- Recipient
- URL Analyser
- Domain checker
- Feature Extractor
- ML Classifier
- Threat Score
- Alert
- Quarantine Manager

Unique Feature:

Each email receives a dynamic threat score such as:

Threat Score = Sender Risk + URL Risk + Content Risk + Domain Risk

High-risk emails can automatically be moved to quarantine.

5. SMART DIGITAL EVIDENCE TAMPERING DETECTION SYSTEM

REAL-TIME CASE STUDY: DIGITAL FORENSIC LABORATORIES.

Problem:

Digital evidence may be accidentally or intentionally modified after collection.

Novelty:

A system that continuously verifies evidence integrity instead of checking the hash only once during collection.

Main Classes:

- Evidence
- Evidence Repository
- Hash Manager
- Integrity Monitor
- Investigator
- Access Log
- Alert Manager
- Audit Trail

Unique Feature:

The system periodically recalculates the evidence hash.

Original Hash $\neq$ Current Hash $\rightarrow$ Possible Tampering $\rightarrow$ Generate Alert

It can also maintain a complete access history showing which investigator accessed or transferred the evidence.